

MAT 330 Proof Portfolio

Fall 2024 Semester

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1 Chapter 1

Exercise 10

Author: Lesley Wiglesworth

Editor: Name

Proof. Let a and b be positive integers and let $d = \gcd(a, b)$ and $m = \text{lcm}(a, b)$. We will first prove that if t divides both a and b , then t divides d . Consider an integer t that divides both a and b . Then, by definition of divides, there exist integers x_1, x_2 such that $a = tx_1$ and $b = tx_2$. Furthermore, because the greatest common divisor of two integers can be written as a linear combination of those integers, there exist integers y_1 and y_2 such that $d = ay_1 + by_2$. Substituting tx_1 for a and tx_2 for b , we can re-write this linear combination as $d = tx_1y_1 + tx_2y_2$. Factoring out a t from the left side of the equation results in $d = t(x_1y_1 + x_2y_2)$. Since $x_1, x_2, y_1, y_2 \in \mathbb{Z}$, $x_1y_1 + x_2y_2 = z \in \mathbb{Z}$. Therefore, $d = t \cdot z$, and thus t must also divide d .

Now we will prove that if s is a multiple of both a and b , then s is a multiple of $m = \text{lcm}(a, b)$. Observe that a and b divide both m and s . Now, we must show that m divides s . If $m = s$, then clearly $m|s$. Suppose that $m \neq s$ (then by definition $m < s$). It follows from the division algorithm that there exist $q, r \in \mathbb{Z}$ such that $s = mq + r$, where $0 \leq r < m$. [Show $r = 0$.] Because a and b both divide s and m , a and b must also divide $r = s - mq$. Therefore, because $0 \leq r < m$ and $m = \text{lcm}(a, b)$, r must equal 0. $\square$

Exercise 38 Prove that for every integer n , $n^3 \pmod{6} = n \pmod{6}$.

Author: Johnson Subedi

Editor: Name

Proof. Let n be integer. Consider the expression $n^3 - n$. Observe that $n^3 - n = n(n+1)(n-1)$ and is therefore the product of 3 consecutive integers. Therefore $n^3 - n$ is divisible by both 2 and 3. Since it is divisible by 2 and 3, it is also divisible by 6. Therefore, $n^3 - n = 0 \pmod{6}$ which implies that $n^3 \pmod{6} = n \pmod{6}$. $\square$

2 Chapter 2

Exercise 19 Prove that the set of all 2×2 matrices with entries from $\mathbf{R}$ and determinant $+1$ is a group under matrix multiplication.

Author: Ben Marrillia

Editor: Field Crumley

Proof. To prove that this set is a group, we need to show that it is closed, associative, has an identity element, and that each element has an inverse. Also assume that all lowercase letters below are arbitrary real numbers.

- Closure: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$, with nonzero determinants.

To show that the set is closed, it must be shown that the product of A and B is a 2×2 matrix with nonzero determinants. It is known that AB is a 2×2 matrix because A and B are 2×2 matrices, and the product of any two matrices with dimensions $m \times n$ and $n \times p$ has dimensions $m \times p$. Furthermore, it is known that $\det(A) \neq 0$, and $\det(B) \neq 0$. When combined with the fact that $\det(A)\det(B) = \det(AB)$, AB must have a nonzero determinant. Therefore, AB is in the set, so the set is closed.

- Associative: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$, and $C = \begin{bmatrix} i & j \\ k & l \end{bmatrix}$.

To show that the set is associative, we will show that $(AB)C = A(BC)$.

$$\begin{aligned}
(AB)C &= \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix} \times \begin{bmatrix} i & j \\ k & l \end{bmatrix} \\
&= \begin{bmatrix} (ae + bg)i + (af + bh)k & (ae + bg)j + (af + bh)l \\ (ce + dg)i + (cf + dh)k & (ce + dg)j + (cf + dh)l \end{bmatrix} \\
&= \begin{bmatrix} aei + bgi + afk + bhk & aej + bgj + afl + bhl \\ cei + dgi + cfk + dhk & cej + dgj + cfl + dhl \end{bmatrix}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
A(BC) &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} ei + fk & ej + fl \\ gi + hk & gj + hl \end{bmatrix} \\
&= \begin{bmatrix} a(ei + fk) + b(gi + hk) & a(ej + fl) + b(gj + hl) \\ c(ei + fk) + d(gi + hk) & c(ej + fl) + d(gj + hl) \end{bmatrix} \\
&= \begin{bmatrix} aei + afk + bgi + bhk & aej + afl + bgj + bhl \\ cei + cfk + dgi + dhk & cej + cfl + dgj + dhl \end{bmatrix}
\end{aligned}$$

Therefore, $(AB)C = A(BC)$, so the set is associative.

- Identity: To show that the set has an identity element, prove that

$Ae = A = eA$. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then

$$Ae = \begin{bmatrix} a(1) + b(0) & a(0) + b(1) \\ c(1) + d(0) & c(0) + d(1) \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A.$$

$$\text{Likewise, } eA = \begin{bmatrix} 1(a) + 0(c) & 1(b) + 0(d) \\ 0(a) + 1(c) & 0(b) + 1(d) \end{bmatrix} = A.$$

Therefore, $Ae = A = eA$, so e is the identity element of the set of 2×2 identity matrices, under multiplication, with determinant $+1$.

- Inverse: To show that each element in the set has an inverse, we need to prove that for any element A , there exists an element A' such that $A(A') = e = A'(A)$.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and let $A' = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$, where $(ad - bc) \neq 0$ because $\det(A) \neq 0$.

Then,

$$\begin{aligned}
A \times A' &= \begin{bmatrix} a(\frac{d}{ad-bc}) + b(\frac{-c}{ad-bc}) & a(\frac{-b}{ad-bc}) + b(\frac{a}{ad-bc}) \\ c(\frac{d}{ad-bc}) + d(\frac{-c}{ad-bc}) & c(\frac{-b}{ad-bc}) + d(\frac{a}{ad-bc}) \end{bmatrix} \\
&= \begin{bmatrix} \frac{ad-bc}{ad-bc} & \frac{-ab+ab}{ad-bc} \\ \frac{-ac+ac}{ad-bc} & \frac{-bc+ad}{ad-bc} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
&= e
\end{aligned}$$

Similarly,

$$\begin{aligned}
A' \times A &= \begin{bmatrix} \frac{d}{ad-bc}(a) - \frac{b}{ad-bc}(c) & \frac{d}{ad-bc}(b) - \frac{b}{ad-bc}(d) \\ \frac{-c}{ad-bc}(a) + \frac{a}{ad-bc}(c) & \frac{-c}{ad-bc}(b) + \frac{a}{ad-bc}(d) \end{bmatrix} \\
&= \begin{bmatrix} \frac{ad-bc}{ad-bc} & \frac{-ab+ab}{ad-bc} \\ \frac{-ac+ac}{ad-bc} & \frac{-bc+ad}{ad-bc} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
&= e
\end{aligned}$$

Therefore, $A(A') = e = A'(A)$, so the set has an inverse.

Since the set is closed, associative, has an identity, and has an inverse for each element, the set of all 2×2 matrices with entries from $\mathbb{R}$ and determinant $+1$ is a group under matrix multiplication. $\square$

Exercise 25 Prove that a group G is Abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all a and b in G .

Author: Rosemary Mihindukulasuriya

Editor: Shelby Wirth

Proof. $\Rightarrow$ [If G is Abelian, then $(ab)^{-1} = a^{-1}b^{-1}$] Let a and b be elements in G and let G be Abelian. Then $a^{-1}b^{-1} = b^{-1}a^{-1}$ for all a^{-1} and b^{-1} in G . Observe that for groups $(ab)^{-1} = b^{-1}a^{-1}$. Therefore $(ab)^{-1} = a^{-1}b^{-1}$.

$\Leftarrow$ [If $(ab)^{-1} = a^{-1}b^{-1}$, then G is Abelian] We will show that $ab = ba$ for all $a, b \in G$. Let $(ab)^{-1} = a^{-1}b^{-1}$ for all a and b in G . Note that $(ab)^{-1} = b^{-1}a^{-1}$ because G is a group. Then $a^{-1}b^{-1} = b^{-1}a^{-1}$ for all a and b in G . Since $(ab)^{-1} = a^{-1}b^{-1} = (ba)^{-1}$, then $(ab)^{-1} = (ba)^{-1}$. Since the inverses are unique, then $ab = ba$. Therefore G is Abelian. $\square$

Exercise 34 Prove that in a group, $(ab)^2 = a^2b^2$ iff $ab = ba$ and $(ab)^{-2} = a^{-2}b^{-2}$ iff $ab = ba$.

Author: Field Crumley

Editor: Noah Calhoun

Proof. For this proof, we will prove the two statements above,

1. We will first prove that $(ab)^2 = a^2b^2$ iff $ab = ba$.

- Let $(ab)^2 = a^2b^2$. Then:

$$\begin{aligned}(ab)(ab) &= a^2b^2 \\ abab &= a^2b^2\end{aligned}$$

Multiplying on the right by b^{-1} yields:

$$\begin{aligned}ababb^{-1} &= a^2b^2b^{-1} \\ aba &= a^2b\end{aligned}$$

Multiplying both sides on the left by a^{-1} yields:

$$\begin{aligned}a^{-1}aba &= a^{-1}a^2b \\ ab &= ba\end{aligned}$$

Thus $(ab)^2 = a^2b^2 \implies ab = ba$.

- Let $ab = ba$. Then:

$$\begin{aligned}(ab)(ba) &= (ab)(ab) \\ abba &= a(ba)b \\ abba &= aabb \\ ab(ba) &= aabb \\ abab &= aabb \\ (ab)(ab) &= (aa)(bb)\end{aligned}$$

And, therefore,

$$(ab)^2 = a^2b^2$$

Thus $(ab)^2 = a^2b^2$ iff $ab = ba$.

2. We will also prove that $(ab)^{-2} = b^{-2}a^{-2}$ iff $ab = ba$.

- Let $(ab)^{-2} = b^{-2}a^{-2}$. Then:

$$(ab)^{-1}(ab)^{-1} = b^{-2}a^{-2}$$

So,

$$(ab)^{-1}b^{-1}a^{-1} = b^{-2}a^{-2}$$

Multiplying both sides by a on the right,

$$(ab)^{-1}b^{-1}a^{-1}a = b^{-2}a^{-2}a$$

$$(ab)^{-1}b^{-1} = b^{-2}a^{-1}$$

$$(b^{-1}a^{-1})b^{-1} = b^{-2}a^{-1}$$

Multiplying both sides by b ,

$$bb^{-1}a^{-1}b^{-1} = bb^{-2}a^{-1}$$

$$a^{-1}b^{-1} = b^{-1}a^{-1}$$

Meaning that, by the “Socks and Shoes Theorem”,

$$(ba)^{-1} = (ab)^{-1}$$

And hence, by taking the inverse of both sides,

$$ba = ab$$

- Let $ab = ba$. Then $(ab)^{-1} = (ba)^{-1}$, meaning that, by the “Socks and Shoes Theorem”,

$$b^{-1}a^{-1} = a^{-1}b^{-1}$$

Then, by multiplying by b^{-1} on the left and by a^{-1} on the right,

$$b^{-1}b^{-1}a^{-1}a^{-1} = b^{-1}a^{-1}b^{-1}a^{-1}$$

Meaning,

$$b^{-2}a^{-2} = (b^{-1}a^{-1})(b^{-1}a^{-1})$$

$$b^{-2}a^{-2} = (ab)^{-1}(ab)^{-1}$$

$$b^{-2}a^{-2} = (ab)^{-2}$$

Hence,

$$(ab)^{-2} = b^{-2}a^{-2}$$

As such, $(ab)^{-2} = a^{-2}b^{-2}$ iff $ab = ba$.

□

3 Chapter 3

Exercise 20 For any group elements a and b , prove that $|ab| = |ba|$.

Author: Gabriel Eze

Editor: Ben Marrillia

Proof. Let G be any group, let $a, b \in G$, and let $|ab| = n$.

This means:

$$(ab)^n = e$$

Multiplying both sides by a^{-1} on the left and a on the right yields:

$$a^{-1}(ab)^na = a^{-1}ea$$

This simplifies to:

$$a^{-1}(ab)^na = (a^{-1}a)e$$

$$a^{-1}(ab)^na = ee$$

$$a^{-1}(ab)^na = e$$

Since $(ab)^n = ab \cdot ab \cdot \dots \cdot ab$ (n times), we can rewrite the equation as:

$$\begin{aligned}
a^{-1}(abab \dots ab)a &= e \\
a^{-1}a(baba \dots ba) &= e \\
e(baba \dots ba) &= e \\
(baba \dots ba) &= e \\
(ba)^n &= e
\end{aligned}$$

This means $(ab)^n = (ba)^n$.
Therefore, $|ab| = |ba|$.

□

Exercise 34 If H and K are subgroups of G , show that $H \cap K$ is a subgroup of G .

Author: Ben Wafford

Editor: Rosemary M

Proof. We will use the two-step subgroup test to prove this.

- Let H and K both be subgroups of G . Since H and K are both subgroups, $e \in H$ and $e \in K$, which implies that $e \in H \cap K$. Therefore $H \cap K$ is non-empty.
- Assume $x, y \in H \cap K$. [Show $xy \in H \cap K$] This means $x, y \in H$ and $x, y \in K$. Since H is a subgroup, we know $xy \in H$ and similarly, since K is a subgroup, $xy \in K$. Therefore, $xy \in H \cap K$.
- Assume $a \in H \cap K$. [Show $a^{-1} \in H \cap K$] This means $a \in H$ and $a \in K$. Since H is a subgroup, $a^{-1} \in H$ and similarly, since K is a subgroup, $a^{-1} \in K$. Therefore, $a^{-1} \in H \cap K$.

Therefore, by the two-step subgroup test, $H \cap K \leq G$.

□

Exercise 38

Let G be an Abelian group and $H = \{x \in G \mid |x| \text{ is odd}\}$.

Prove that H is a subgroup of G .

Author: Shelby Wirth

Editor: Gabriel Eze

Proof. Let G be an Abelian group and define $H = \{x \in G \mid |x| \text{ is odd}\}$.

We will show that H is a subgroup of G by verifying the following properties:

- **Non-empty:** Since $|e| = 1$, which is odd, the identity element $e \in H$.
- **Closure:** Let $a, b \in H$, meaning $|a| = m$ and $|b| = n$ where both m and n are odd.

We know that $|ab|$ must divide $\text{lcm}(m, n) = k$. Since k is odd and $|ab|$ divides k ; we have $|ab|$ to be odd. Therefore, $ab \in H$ and H is closed under the group operation.

- **Inverses:** Let $a \in H$, which means $|a| = m$, where m is odd. We apply the property of orders:

$$|a^{-1}| = |a| = m$$

Thus, $a^{-1} \in H$ and H is closed under inverses.

Therefore, H is an Abelian subgroup of G by 2SST. □

Exercise 47

Let G be an Abelian group with identity e and let n be some fixed integer. Prove that the set of all elements of G that satisfy the equation $x^n = e$ is a subgroup of G . Give an example of a group G in which the set of all elements of G that satisfy the equation $x^2 = e$ does not form a subgroup of G .

Author: Drew Davidson

Editor: Ben Wafford

Proof. Let G be an Abelian group with identity e and let n be some fixed integer. Then, $H = \{x \in G \mid x^n = e\}$ is a subgroup of G . From here, we will proceed with the Two-Step Subgroup Test.

- H is nonempty since $x^n = e \in H$.
- Let $a, b \in H$. Then, $a^n = e$ and $b^n = e$. Then:

$$\begin{aligned} (ab)^n &= abab\dots ab \\ &= a^n b^n \\ &= ee \\ &= e \end{aligned}$$

- Let $a \in H$. Then, $a^n = e$. Therefore, $(a^{-1})^n = (a^n)^{-1} = e^{-1} = e$. So, $a^{-1} \in H$.

Therefore, by the Two-Step Subgroup Test, the set of all elements of G that satisfy the equation $x^n = e$ is a subgroup of G . □

An example of a group G in which the set of all elements of G that satisfy the equation $x^2 = e$ does not form a subgroup of G is D_4 .

4 Chapter 4

Exercise 15

Let G be an Abelian group and let $H = \{g \in G \mid |g| \text{ divides } 12\}$. Prove that H is a subgroup of G . Is there anything special about 12 here? Would your proof be valid if 12 were replaced by some other positive integer? State the general result.

Author: Johnson Subedi

Editor: Andrew Kahn

Proof. Let G be an Abelian group and let $H = \{g \in G \mid |g| \text{ divides } 12\}$. To prove H is a subgroup of G , we need to show:

- **H is non-empty:** Since G is a group, we know the identity element $e \in G$. $|e| = 1$ which divides 12. Therefore $e \in H$.
- **H is closed under the group operation:** Let $a, b \in H$. Then $|a|$ and $|b|$ divide 12. Let $\text{lcm}(|a|, |b|) = k$. Then we know k divides 12 since 12 is divisible by both $|a|$ and $|b|$. Now, $|ab|$ divides $\text{lcm}(|a|, |b|) = k$ (a property of Abelian groups). Therefore, $|ab|$ divides 12, so $ab \in H$.
- **H contains inverses of its elements:** If $a \in H$, then $|a|$ divides 12. Note that $|a^{-1}| = |a|$. So $|a^{-1}|$ also divides 12. Thus, $a^{-1} \in H$.

Therefore, by the two step subgroup test, H is a subgroup of G .

Generalization: There is nothing special about 12. The proof would be valid for any positive integer n . We can state the general result as: For any Abelian group G and any positive integer n , the set $H = \{g \in G \mid |g| \text{ divides } n\}$ is a subgroup of G . □

Exercise 22

Prove that a group of order 3 must be cyclic.

Author: Noah Calhoun

Editor: Drew Davidson

Proof. Let G be a group with order 3. This means $G = \{e, a, b\}$ where e is the identity element. Assume for the sake of contradiction that $aa = e$. What value would we then assign to ba ? If $ba = e = aa$, we know that $b = a$ by right cancellation. Similarly, if $ba = a = ea$, then we know that $b = e$. If we let $ba = b$, then left cancellation indicates that $a = e$. These three scenarios contradict the fact that G is a three element group, so we know that $|a| \neq 2$. We also know that $|a| \neq 1$, otherwise a would be the identity. As the order of an element is less than or equal to the order of the group, $|a| = 3$. Since $|a| = 3$ and $|G| = 3$, we know $\langle a \rangle = G$. $\square$

5 Chapter 5

Exercise 19 Let α and β belong to S_n . Prove that $\alpha\beta$ is even if and only if α and β are both even or both odd.

Author: Henry Davis

Editor: Johnson

Proof. Let α and β be in S_n . We know that each even permutation can be written as the product of even number of transpositions. So let's assume α and β are even transpositions. Thus when taking the product of $\alpha\beta$ we're taking the product of even number of transpositions. Therefore the sum of the 2 cycles can be written as the sum of two evens giving an even. Algebraically looking like;

$$\alpha = (x_1 x_2 \dots x_k) \text{ and } \beta = (y_1 y_2 \dots y_p)$$

$$\alpha\beta = (x_1 x_2 \dots x_k) (y_1 y_2 \dots y_p)$$

Thus the total number of transpositions of $\alpha\beta$ is equal to $k+p$ which we know an even plus an even is even. Therefore when taking the product of two even or two odd permutations you get an even. W.l.o.g the same happens for odd permutations

Going the other way we need to use the parity principle. From the parity principle we know that if the product of $\alpha\beta$ is odd then either α is odd or β is odd. So assuming $\alpha\beta$ is even we know that either α and β both have to be odd or both have to be even.

□

Exercise 45

For $n \geq 3$, let $H = \{\beta \in S_n \mid \beta(1) = 1 \text{ or } 2 \text{ and } \beta(2) = 1 \text{ or } 2\}$.

Prove that H is a subgroup of S_n . Determine $|H|$.

Author: Andrew Kahn

Editor: Henry Davis

Proof. Let S_n be a symmetric group under function composition. To show that H is a subgroup of S_n we need to show:

- **H is non empty:** Since S_n is a group there exists an identity element e . Since $e(1) = 1$ and $e(2) = 2$ we know that $e \in H$ by definition of H .
- **H is closed under the operation of the group, function composition:** Let α and $\beta \in H$. Consider the possible values for:
 - $\alpha(1) = 1 \text{ or } 2$
 - $\alpha(2) = 1 \text{ or } 2$
 - $\beta(1) = 1 \text{ or } 2$
 - $\beta(2) = 1 \text{ or } 2$

The possible compositions are $\alpha\beta(1)$, $\alpha\beta(2)$, $\beta\alpha(1)$, or $\beta\alpha(2)$. All of these possible compositions result is either 1 or 2 by the definition of the subgroup, therefore the group is closed under function composition.

Therefore H is a subgroup of S_n by the finite subgroup test.

□

The order of H is $2 \times 1 \times (n-2)!$

6 Chapter 6

Exercise 31 If ϕ and γ are isomorphisms from the cyclic group $\langle a \rangle$ to some group and $\phi(a) = \gamma(a)$, prove that $\phi = \gamma$.

Author: Rosemary M

Editor: Noah C

Proof. Let ϕ and γ be isomorphisms from the cyclic group $\langle a \rangle$ to some group G and let $\phi(a) = \gamma(a)$. Now let b be an arbitrary element in $\langle a \rangle$. Since $\langle a \rangle$ is cyclic, $b = a^n$ for some positive integer n . Thus $\phi(b) = \phi(a^n)$. As ϕ is operation preserving, we know that $\phi(a^n) = [\phi(a)]^n$. We also know $\phi(a) = \gamma(a)$ so $[\phi(a)]^n = [\gamma(a)]^n = \gamma(a^n) = \gamma(b)$. So we have shown $\phi(b) = \gamma(b)$ for all $b \in \langle a \rangle$. Therefore $\phi = \gamma$. $\square$

Exercise 34 Prove that if K is a subgroup of a group G , and $\phi : G \rightarrow G'$ is an isomorphism, then $\phi(K) = \{\phi(k) \mid k \in K\}$ is a subgroup of G' .

Author: Gabriel Eze

Editor: Rosemary M

Proof. Let $\phi : G \rightarrow G'$ be an isomorphism of groups, and let K be a subgroup of G . We aim to show that $\phi(K) = \{\phi(k) \mid k \in K\}$ is a subgroup of G' by verifying the subgroup criteria.

1. **Non-emptiness:** Since K is a subgroup of G , it contains the identity element e_G . An isomorphism maps identity to identity, so:

$$\phi(e_G) = e_{G'}.$$

Therefore, $e_{G'} \in \phi(K)$, which means $\phi(K) \neq \emptyset$.

2. **Closure under group operation:** For any elements $\phi(a), \phi(b) \in \phi(K)$, $a, b \in K$. Since K is a subgroup, $ab \in K$. Using the fact that ϕ is operation preserving:

$$\phi(a)\phi(b) = \phi(ab) \in \phi(K).$$

Thus, $\phi(K)$ is closed under the group operation.

3. **Closure under inverses:** For any element $\phi(k) \in \phi(K)$, $k \in K$. Since K is a subgroup, $k^{-1} \in K$. Because isomorphisms preserve inverses:

$$\phi(k^{-1}) = (\phi(k))^{-1} \in \phi(K).$$

Thus, $\phi(K)$ is closed under inverse operation.

Therefore, $\phi(K)$ is a subgroup of G' by the Two-Step Subgroup Test. $\square$

Exercise 47 Suppose g and h induce the same inner automorphism of a group G . Prove that $h^{-1}g \in Z(G)$.

Author: Shelby Wirth

Editor: Johnson Subedi

Proof. Let $x \in G$ then, $gxg^{-1} = h x h^{-1}$ for all $x \in G$. Then, right multiplying by g we get, $gx = h x h^{-1}g$. Next, left multiplying by h^{-1} we get $h^{-1}gx = x h^{-1}g$. Which is equal to $(h^{-1}g)x = x(h^{-1}g)$. Thus, $h^{-1}g$ commutes with all $x \in G$ and by definition $h^{-1}g \in Z(G)$. $\square$

7 Chapter 7

Exercise 26 Suppose that G is a group with more than one element and G has no proper, non-trivial subgroups. Prove that $|G|$ is prime.

Author: Ben Wafford

Editor: Gabriel Eze

Proof. Let G be a group with more than one element. Since G is a group, it must contain an identity element, say e . By the given condition, G has no proper, non-trivial subgroups. This implies that the only subgroups of G are the trivial subgroup $\{e\}$ and the group G itself.

Consider the order of G , denoted by $|G|$. Let $|G| = n$. We want to show that n must be a prime number.

Suppose, for contradiction, that n is not a prime. Then n must be composite, meaning that there exists an integer $1 < d < n$ such that d divides n . By Lagrange's theorem, the order of any subgroup of G must divide the order of G . Thus, there would exist a subgroup of order d , where $1 < d < n$, which would be a proper, non-trivial subgroup of G . This contradicts the assumption that G has no proper, non-trivial subgroups.

Therefore, n must be a prime number. Hence, $|G|$ is prime. $\square$

Exercise 27 Let $|G| = 15$. If G has only one subgroup of order 3 and only one of order 5, prove that G is cyclic. Generalize to $|G| = pq$, where p and q are prime.

Author: Ben Marrillia

Editor: Drew Davidson

Proof. Let H and K be the subgroups of G with orders 3 and 5, respectively. From Lagrange's Theorem, we know that any group of prime order is cyclic. So, each nonidentity $h \in H$ has order 3 and each nonidentity $k \in K$ has order 5 because the order of any group element must divide the order of the group. Then, the only element in $H \cap K$ is the identity. Therefore, $H \cup K$ has 7 elements, which is less than 15. Thus, there exists $g \in G$ that is not in $H \cup K$.

From here, we know that $g \neq e \in G$ and that $|g|$ must divide 15, so $|g| = 3, 5$ or 15 . However, if $|g| = 3$, then $\langle g \rangle = H$, and we already determined $g \notin H$. Similarly, if $|g| = 5$, $\langle g \rangle = K$. Therefore, $|g| = 15 = |G|$, which means G is cyclic.

In general terms, for any group G whose order is the product of two primes p and q , $H \cup K$ has $p + q - 1 < pq$ elements, so there is always a $g \in G$ where $g \notin H \cup K$. Therefore, $|g| = p, q$, or pq because $|g|$ must divide $|G| = pq$. However, $|g| \neq p$ or q because the cyclic groups generated by these elements would be H or K , respectively. So, $|g| = pq = |G|$, meaning G is cyclic. $\square$

Exercise 38 Prove that if G is a finite group, the index of $Z(G)$ cannot be prime.

Author: Noah Calhoun

Editor: Ben Wafford

Proof. Let G be a finite group and let $Z(G)$ be the center of G . If G is Abelian, then $[G : Z(G)] = 1$, and thus non-prime. If G is non-Abelian then there exists an element $a \in G$ where $a \notin Z(G)$. Consider the set $K = \langle a \rangle Z(G)$. We know that $H < K$ as $e \in \langle a \rangle$. We also know that $K \leq G$ as it is closed under multiplication and contains an inverse for every element. We also know $K \neq G$ because K is Abelian, the powers of a commute with themselves and the elements of $Z(G)$ commute with every element, and G is non-Abelian. Thus $H < K < G$.

Let $|H| = h$, as $H < K$ by Lagrange's theorem we know that h divides $|K|$. As H is a subgroup of G we know that h divides $|G|$. As such $|G| = hp$ and $|K| = hq$ for some $p, q \in \mathbb{Z}^+$. We also know that $K < G$ so, by Lagrange's theorem, $|K|$ must divide $|G|$. Thus $(hq)|(hp) \implies q|p$. As such p cannot be prime, $\frac{|G|}{|H|} = \frac{hp}{h} = p$, thus $[G : H] \neq p$. $\square$

Exercise 41 Let G be a group of order 100 that has a subgroup H of order 25. Prove that every element of G of order 5 is in H .

Author: Drew Davidson

Editor: Henry Davis

Proof. Let g be an element with order 5 in G . Using *Theorem 7.2*, we know the following:

$$|\langle g \rangle H| = \frac{|\langle g \rangle| |H|}{|\langle g \rangle \cap H|} = \frac{5 \cdot 25}{|\langle g \rangle \cap H|} = \frac{125}{|\langle g \rangle \cap H|}$$

We also know that $|G| = 100$, which makes $|\langle g \rangle H|$ no more than 100 because it is a subset of G . This then forces $|\langle g \rangle \cap H|$ to be greater than 1. Next, we can use *Lagrange's Theorem* to find the exact order of $|\langle g \rangle \cap H|$. Because $|\langle g \rangle|$ is prime, $|\langle g \rangle \cap H|$ must be 5. Therefore every element of G of order 5 is in H . $\square$

8 Chapter 8

Exercise 37 Let $G = \{3^m 6^n | m, n \in \mathbb{Z}\}$ under multiplication. Prove that G is isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$. Does your proof remain valid if $G = \{3^m 9^n | m, n \in \mathbb{Z}\}$?

Author: Andrew Kahn

Editor: Ben Marrillia

Proof. Let $G = \{3^m 6^n | m, n \in \mathbb{Z}\}$ be a group under multiplication. To show that G is isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$, we first define a mapping $\phi(3^m 6^n) = (m, n)$. Now we show the mapping is:

1 – 1: Let $\phi(3^m 6^n) = \phi(3^a 6^b)$, where $3^m 6^n$ and $3^a 6^b \in G$ and $a, b \in \mathbb{Z}$. This implies that $(m, n) = (a, b)$ which then implies that $m = a$ and $n = b$. Thus, ϕ is 1-1.

Onto: Let $\phi(3^m 6^n) = (m, n)$. Since m and n are arbitrary elements of $\mathbb{Z}$, we see ϕ is onto.

Operation preserving: Observe that, for $a, b, m, n \in Z$,

$$\begin{aligned}\phi((3^m 6^n)(3^a 6^b)) &= (3^{(m+a)} 6^{(n+b)}) \\ &= (m+a, n+b) \\ &= (m, n)(a, b) \\ &= \phi(3^m 6^n) \phi(3^a 6^b).\end{aligned}$$

Thus, ϕ is operation preserving.

Therefore, G is isomorphic to $Z \oplus Z$ by the definition of an isomorphism. $\square$

If $G = \{3^m 9^n | m, n \in Z\}$, our proof is not valid since the mapping is not well-defined. For example, $3^2 9^0 = 3^0 9^1$, but $\phi(3^2 9^0) = (2, 0) \neq (0, 1) = \phi(3^0 9^1)$.

Exercise 59: Let p be prime. Prove that $\mathbb{Z}_p \oplus \mathbb{Z}_p$ has exactly $p+1$ subgroups of order p .

Author: Johnson Subedi

Editor: Andrew Kahn

Proof. Since $\mathbb{Z}_p \oplus \mathbb{Z}_p$ is cyclic, all elements have order p or 1. The only element of order 1 is the identity, and there are $|\mathbb{Z}_p \oplus \mathbb{Z}_p| - 1 = p^2 - 1$ elements of order p . If G is a subgroup of order p , then it is cyclic. So there are $p-1$ elements of order p in G . Furthermore, if G and H are two distinct subgroups of order p , then $|G \cap H| \mid |G| = p$ and $|G \cap H| \neq p$. So $G \cap H = \{e\}$.

Therefore if we denote the number of subgroups with order p by k , then the number of elements of order p in G is $(p-1)k$. Since $(p-1)k = p^2 - 1 = (p-1)(p+1)$, we know that $k = p+1$. Thus $\mathbb{Z}_p \oplus \mathbb{Z}_p$ has exactly $p+1$ subgroups of order p . $\square$

9 Chapter 9

Exercise 11 Prove that a factor group of a cyclic group is cyclic.

Author: Field Crumley

Editor: Shelby Wirth

Proof. Recall the definition of a factor group H : $G/H = \{aH : a \in G\}$. Since G is cyclic, there is some element $a \in G$ such that $G = \langle a \rangle$. Then,

let H be normal in G . By definition, any element of G/H has the form a^kH . Which can be written as $(aH)^k$ for some $k \in G$. This means $G/H = \langle aH \rangle$ and is cyclic. Therefore, a factor group of a cyclic group is cyclic. $\square$

Exercise 39 Let H be a subgroup of a group G with the property that for all a and b in G , $aHbH = abH$. Prove that H is a normal subgroup of G .

Author: Henry Davis

Editor: Field Crumley

Proof. Let g be an arbitrary element in G . We know in order to be a normal subgroup the following property needs to hold: $gHg^{-1} \subseteq H$.

Since $aHbH = abH$ is true for all elements in G , we can let $a = g$ and $b = g^{-1}$. This produces the value

$$gHg^{-1}H = gg^{-1}H \tag{1}$$

$$gHg^{-1}H = eH \tag{2}$$

$$gHg^{-1}H = H \tag{3}$$

. Note that $g \in gH$ (first property of cosets, p. 139). Then, for an arbitrary $h \in H$, we can say $gh = l$, meaning

$$lg^{-1}h = H \tag{4}$$

Meaning $lg^{-1} \in H$, by the second property of cosets. Thus,

$$lg^{-1} \in H \tag{5}$$

$$ghg^{-1} \in H \tag{6}$$

Hence, because g, g^{-1}, h and l were arbitrary, $gHg^{-1} \subseteq H$. Therefore, by the Normal Subgroup Test, H is a normal subgroup of G . $\square$

10 Chapter 10